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Employment	Assistant Professor, Halicioğlu Data Science Institute, University of California, San Diego	2025 – present
	Research Assistant Professor, Toyota Technological Institute at Chicago	2024 – 2025
Education	Ph.D. in Statistics and Data Science, Yale University	2024
	• Advisor: Prof. Zhou Fan	
	M.A. in Statistics (en route), Yale University	2021
	B.S. in Mathematics, University of Science and Technology of China (USTC)	2014 – 2018
	• B.E. in Computer Science (dual)	
Publications and Preprints	(*: equal contribution)	
	1. Shuo Xie*, Tianhao Wang*, Beining Wu, and Zhiyuan Li. “A Tale of Two Geometries: Adaptive Optimizers and Non-Euclidean Descent”. Available at arXiv: 2511.20584.	
	2. Mohamad Amin Mohamadi, Tianhao Wang, and Zhiyuan Li. “Honesty over Accuracy: Trustworthy Language Models through Reinforced Hesitation”. Available at arXiv: 2511.11500.	
	3. Xingjian Zhang, Tianhong Gao, Suliang Jin, Tianhao Wang, Teng Ye, Eytan Adar, and Qiaozhu Mei. “Through the Judge’s Eyes: Inferred Thinking Traces Improve Reliability of LLM Raters”. Available at arXiv: 2510.25860.	
	4. Robin Yadav, Shuo Xie, Tianhao Wang, and Zhiyuan Li. “Provable Benefit of Sign Descent: A Minimal Model Under Heavy-Tail Class Imbalance”. NeurIPS 2025 Workshop on Optimization for Machine Learning (OPT 2025). (Oral) .	
	5. Max Lovig, Tianhao Wang, and Zhou Fan. “On Universality of Non-Separable Approximate Message Passing Algorithms”. Available at arXiv: 2506.23010.	
	6. Siyu Chen, Heejune Sheen, Xuyuan Xiong, Tianhao Wang, and Zhuoran Yang. “Taming Polysemanticity in LLMs: Provable Feature Recovery via Sparse Autoencoders”. Available at arXiv: 2506.14002.	
	7. Shuo Xie, Tianhao Wang, Sashank Reddi, Sanjiv Kumar, and Zhiyuan Li. “Structured Preconditioners in Adaptive Optimization: A Unified Analysis”. <i>International Conference on Machine Learning (ICML)</i> , 2025. Available at arXiv: 2503.10537.	
	8. Siyu Chen*, Beining Wu*, Miao Lu, Zhuoran Yang, and Tianhao Wang. “Can Neural Networks Achieve Optimal Computational-statistical Tradeoff? An Analysis on Single-Index Model”. <i>International Conference on Learning Representations (ICLR)</i> , 2025 (Oral) .	
	9. Angeliki Giannou, Liu Yang, Tianhao Wang, Dimitris Papailiopoulos, and Jason D. Lee. “How well can Transformers emulate in-context Newton’s method?” <i>International Conference on Artificial Intelligence and Statistics (AISTATS)</i> , 2025. Available at arXiv:2403.03183.	
	10. Siyu Chen*, Heejune Sheen*, Tianhao Wang, and Zhuoran Yang. “Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers”. <i>Annual Conference on Neural Information Processing Systems (NeurIPS)</i> , 2024. Available at arXiv:2409.10559.	
	11. Xinyi Zhong*, Tianhao Wang*, and Zhou Fan. “Approximate Message Passing for orthogonally invariant ensembles: Multivariate non-linearities and spectral initialization”. <i>Information and Inference</i> , 2024. Available at arXiv:2110.02318.	
	12. Tianhao Wang, Xinyi Zhong, and Zhou Fan. “Universality of Approximate Message Passing algorithms and tensor networks”. <i>Annals of Applied Probability</i> , 2024. Available at arXiv:2206.13037.	
	13. Heejune Sheen, Siyu Chen, Tianhao Wang, and Harrison H. Zhou. “Implicit Regularization of Gradient Flow on One-Layer Softmax Attention”. Available at arXiv:2403.08699.	

14. Siyu Chen, Heejune Sheen, Tianhao Wang, and Zhuoran Yang. “Training dynamics of multi-head softmax attention for in-context learning: emergence, convergence, and optimality”. *Conference on Learning Theory (COLT)*, 2024. Available at arXiv:2402.19442.
15. Zhou Fan, Roy R. Lederman, Yi Sun, Tianhao Wang, Sheng Xu. “Maximum likelihood for high-noise group orbit estimation and single-particle cryo-EM”. *Annals of Statistics*, 2024. Available at arXiv:2107.01305.
16. Runzhe Wang, Sadhika Malladi, Tianhao Wang, Kaifeng Lyu, and Zhiyuan Li. “The Marginal Value of Momentum for Small Learning Rate SGD”. *International Conference on Learning Representations (ICLR)*, 2024. Available at arXiv:2307.15196.
17. Zhou Fan, Yi Sun, Tianhao Wang, Yihong Wu. “Likelihood landscape and maximum likelihood estimation for the discrete orbit recovery model”. *Communications on Pure and Applied Mathematics*, 2023. Available at arXiv:2004.00041.
18. Ruitu Xu, Yifei Min, and Tianhao Wang. “Noise-adaptive Thompson sampling for linear contextual bandits”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
19. Yifei Min, Jiafan He, Tianhao Wang, Quanquan Gu. “Cooperative multi-agent reinforcement learning: Asynchronous communication and linear function approximation”. *International Conference on Machine Learning (ICML)*, 2023. Available at arXiv:2305.06446.
20. Ruitu Xu, Yifei Min, Tianhao Wang, Michael I. Jordan, Zhaoran Wang, Zhuoran Yang. “Finding regularized competitive equilibria of heterogeneous agent macroeconomic models via reinforcement learning”. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023. Available at arXiv:2303.04833.
21. Zhiyuan Li, Tianhao Wang, Dingli Yu. “Fast mixing of stochastic gradient descent with normalization and weight decay”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022.
22. Zhiyuan Li*, Tianhao Wang*, Jason D. Lee, Sanjeev Arora. “Implicit bias of gradient descent on reparametrized models: on equivalence to mirror descent”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022. Available at arXiv:2207.04036.
23. Jiafan He*, Tianhao Wang*, Yifei Min*, Quanquan Gu. “A simple and provably efficient algorithm for asynchronous federated linear bandits”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022. Available at arXiv:2207.03106.
24. Yifei Min, Tianhao Wang, Ruitu Xu, Zhaoran Wang, Michael I. Jordan, Zhuoran Yang. “Learn to match with no regret: Reinforcement learning in Markov matching market”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022 (**Oral**). Available at arXiv:2203.03684.
25. Yifei Min, Jiafan He, Tianhao Wang, Quanquan Gu. “Learning stochastic shortest path with linear function approximation”. *International Conference on Machine Learning (ICML)*, 2022. Available at arXiv:2110.12727.
26. Zhiyuan Li, Tianhao Wang, Sanjeev Arora. “What happens after SGD reaches zero loss? – A mathematical framework”. *International Conference on Learning Representations (ICLR)*, 2022 (**Spotlight**). Available at arXiv:2110.06914.
27. Pamela L Valentino, Tianhao Wang, Veronika Shabanova, Vicky Lee Ng, John C Bucuvalas, Amy G Feldman, Regino P Gonzalez-Peralta, Nitika Arora Gupta, Tamir A Miloh, Saeed Mohammad, Erika Pace, Shikha S Sundaram, Nada A Yazigi, Kyle Soltys, Society of Pediatric Liver Transplantation (SPLIT). “North American biliary stricture management strategies in children post liver transplant: multicenter analysis from the SPLIT Registry”. *Liver Transplantation*, 2021.
28. Yifei Min*, Tianhao Wang*, Dongruo Zhou, Quanquan Gu. “Variance-aware off-policy evaluation with linear function approximation”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2021. Available at arXiv:2106.11960.
29. Tianhao Wang*, Dongruo Zhou*, Quanquan Gu. “Provably efficient reinforcement learning with linear function approximation under adaptivity constraints”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2021. Available at arXiv:2101.02195.
30. Pan Xu*, Tianhao Wang*, Quanquan Gu. “Continuous and discrete-time accelerated stochastic mirror descent for strongly convex functions”. *International Conference on Machine Learning (ICML)*, 2018.
31. Pan Xu*, Tianhao Wang*, Quanquan Gu. “Accelerated stochastic mirror descent: From continuous-time dynamics to discrete-time algorithms”. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2018.

Honors and Awards	• Leonard F. Savage Prize for best written work	Yale, 2024
	• Student Poster Competition Award	Rutgers University, 2023
	– At Conference on Recent Advances in Statistics and Data Science	
	• NeurIPS 2022 top reviewer	NeurIPS, 2022
	• Wedworth W. Clarke Fellowship	Yale, 2021
Invited Talks and Seminars	• International Conference on Continuous Optimization, Los Angeles	07/2025
	• Midwest Machine Learning Symposium, Chicago	06/2025
	• International Indian Statistical Association Conference, Lincoln	06/2025
	• Midwest Optimization & Statistical Learning Conference, Chicago	05/2025
	• Flatiron Institute, CCM ML seminar	04/2025
	• Research at TTIC	11/2024
	• FAI seminar (online)	08/2024
	• Joint Statistical Meetings, Portland	08/2024
	• University of Notre Dame, ACMS Colloquium	02/2024
	• UIUC, Statistics Seminar	02/2024
	• Columbia University, Statistics Seminar	01/2024
	• UCSD, Halicioğlu Data Science Institute, Special Seminar Series	01/2024
	• UC Davis, Statistics Seminar	01/2024
	• UCSD, Department of Mathematics Colloquium	12/2023
	• INFORMS Annual Meeting, Phoenix	10/2023
Teaching	Instructor, University of California, San Diego	
	• DSC 242: High-dimensional Probability and Statistics	Fall 2025
	Teaching assistant, Yale University	
	• High-Dimensional Phenomena in Statistics and Learning	Spring 2023
	• Intermediate Machine Learning	Spring 2022
	• Statistical Inference	Fall 2020, Fall 2021
	• Information Theory	Spring 2021
	• Probability and Statistics	Fall 2019
	• Stochastic Processes	Spring 2019, Spring 2020
Professional Service	Organizing committee for	
	• Midwest Optimization & Statistical Learning Conference 2025	
	Conference area chair for	
	• NeurIPS 2025	
	• ICLR 2025	
	Journal reviewer for	
	• Annals of Statistics	
	• IEEE Transactions on Information Theory	

- Journal of the American Statistical Association
- Journal of Machine Learning Research
- Journal of the Royal Statistical Society: Series B
- Random Matrices: Theory and Applications
- SIAM Journal on Mathematics of Data Science
- Statistical Science

Conference reviewer for

- AAAI 2025
- AISTATS 2022, 2023, 2024
- COLT 2025
- ICLR 2023, 2024, 2025
- ICML 2022, 2023, 2024, 2025
- IEEE ITW 2023
- NeurIPS 2022, 2023, 2024
- Reinforcement Learning Conference 2024, 2025